Next-Day Precipitation Prediction at Schiphol

A Gaussian Naive Bayes Approach

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Abstract

This project investigates whether next-day precipitation at Schiphol, the Netherlands, can be predicted from same-day meteorological observations. Daily weather records from KNMI station 240 spanning 1 January 1971 to 28 March 2026 (approximately 20,176 days) were used. Nine base meteorological variables, including sea-level pressure, cloud cover, relative humidity, wind speed, wind direction, visibility, temperature, and sunshine duration were selected as predictors, alongside their day-to-day differences (delta features), yielding 18 features in total. A binary rain label was defined as daily precipitation exceeding 1 mm. A Gaussian Naive Bayes classifier was trained using maximum likelihood estimation on the first 80% of the chronologically ordered data and evaluated on the remaining 20%, achieving 70% overall accuracy with a rain recall of 64%. A logistic regression classifier was used as a benchmark and achieved a 68% overall accuracy and a rain recall of 76%.

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1 Introduction

Precipitation forecasting is a central problem in meteorology and has direct societal relevance for agriculture, hydrology, and urban planning. The Netherlands, situated in a maritime temperate climate, experiences frequent and highly variable precipitation driven by Atlantic weather systems, making short-term rain prediction both scientifically interesting and practically valuable. Precipitation at a given location is governed by synoptic-scale atmospheric dynamics: the passage of frontal systems, variations in sea-level pressure, moisture transport, and atmospheric instability. These processes leave measurable signatures in standard surface observations on the day preceding a rain event, making next-day prediction from same-day observations a meaningful statistical problem.

This project applies a Gaussian Naive Bayes classifier to predict whether the following day will be a rain day (≥ 1 mm of precipitation) at KNMI station Schiphol (station 240), based on 18 meteorological features observed on the current day. Class-conditional likelihoods are estimated using maximum likelihood estimation and combined with Bayes' theorem to make a prediction. The model is compared against a naive baseline and against logistic regression to see how much the independence assumption affects performance.

The report is structured as follows. Section 2 provides the physical background relevant to precipitation formation and the chosen predictors. Section 3 describes the dataset and preprocessing. Section 4 defines the classification models. Section 5 presents the statistical framework. Section 6 describes the inference procedure. Section 7 presents the results. Sections 8 and 9 discuss limitations and possible extensions.

2 Physical Background

Precipitation in the mid-latitudes is primarily associated with the lifting of moist air masses at frontal boundaries and within low-pressure systems. As an air parcel rises, it cools adiabatically; once it reaches the dew point, water vapour condenses to form cloud droplets, which grow through collision and coalescence until precipitation occurs. Sea-level pressure (`PG`) is therefore among the most physically informative predictors: low-pressure systems drive convergence and uplift, while the day-to-day pressure change (`dPG`) captures synoptic development — a falling barometer signals an approaching depression and elevated rain probability. Relative humidity (`UG`), cloud cover (`NG`), sunshine duration (`SQ`), and visibility (`VVN`) all reflect the current moisture and cloud state of the boundary layer. Mean temperature (`TG`) contributes through the Clausius-Clapeyron relation, whereby saturation vapour pressure increases approximately exponentially with temperature. Wind direction (`DDVEC`) is decomposed into sine and cosine components to preserve its circular domain; southwesterly flow in the Netherlands is associated with Atlantic moisture advection [1, 2].

3 Data

3.1 Source

The dataset was obtained from the Royal Netherlands Meteorological Institute (KNMI) for station 240 (Schiphol). The data span from 1 January 1971 to 28 March 2026, comprising approximately 20,176 daily records. The dataset is publicly available through the KNMI Climate Explorer portal [3].

3.2 Measured Quantities

Nine base variables were selected as predictors on physical grounds (see Table 1). Wind direction (`DDVEC`) was decomposed into sine and cosine components to handle its circular domain, yielding

10 base features. Day-to-day differences (delta features) were computed for each base variable as:

$$\Delta x_t = x_t - x_{t-1} \tag{1}$$

giving 18 features in total. Delta features capture temporal trends (e.g., a falling barometer) that carry predictive information beyond the instantaneous value alone.

Code	Description	Unit
`NG`	Mean daily cloud cover	octants (0–8)
`UG`	Mean relative humidity	%
`PG`	Mean sea-level pressure	hPa
`FG`	Mean wind speed	0.1 m/s
`DDVEC_sin`	Wind direction (sine component)	–
`DDVEC_cos`	Wind direction (cosine component)	–
`VVN`	Minimum visibility	coded scale
`TG`	Mean temperature	0.1 °C
`SQ`	Sunshine duration	0.1 h

Table 1: Base predictor variables selected from the KNMI dataset. Each variable also appears in delta form (`d*`), giving 18 predictors in total.

3.3 Preprocessing

The binary rain label was defined as:

$$y_{t+1} = \begin{cases} 1 & \text{if } \mathrm{RH}_{t+1} \geq 10 \text{ (i.e. } \geq 1\,\mathrm{mm}) \\ 0 & \text{otherwise} \end{cases} \tag{2}$$

where `RH` is stored in units of 0.1 mm in the KNMI file. Labels were shifted by one day relative to the feature rows so that features at day t predict rain at day $t+1$.
Rows with missing values in any predictor or the label were removed. The dataset was split chronologically into 80% training and 20% test sets. The resulting test set contains approximately 4,033 days, of which roughly 35% are rain days.

4 Model

4.1 Gaussian Naive Bayes

The classifier is based on Bayes' theorem. Given a feature vector $\mathbf{x} = (x_1, \ldots, x_d)$ with $d = 18$, the posterior probability of class $c \in \{0, 1\}$ is:

$$P(c \mid \mathbf{x}) = \frac{P(\mathbf{x} \mid c)\, P(c)}{P(\mathbf{x})} \tag{3}$$

Since the denominator $P(\mathbf{x})$ is identical for both classes, classification reduces to:

$$\hat{y} = \arg\max_c \left[\log P(\mathbf{x} \mid c) + \log P(c)\right] \tag{4}$$

Under the Gaussian Naive Bayes assumption, features are conditionally independent given the class, and each feature x_j follows a univariate Gaussian distribution:

$$P(\mathbf{x} \mid c) = \prod_{j=1}^{d} \mathcal{N}(x_j;\, \mu_{jc},\, \sigma_{jc}^2) \tag{5}$$

The log-posterior therefore becomes:

$$\log P(c \mid \mathbf{x}) \propto \sum_{j=1}^{d} \left[-\frac{(x_j - \mu_{jc})^2}{2\sigma_{jc}^2} - \log \sigma_{jc} \right] + \log P(c) \tag{6}$$

The model parameters are $\boldsymbol{\theta} = \{\mu_{jc}, \sigma_{jc}^2\}$ for $j = 1, \ldots, 18$ and $c \in \{0, 1\}$, giving 72 parameters in total, plus the two class priors $P(c)$.
The key assumption is that conditional independence of features given the class is a simplification, as meteorological variables are physically correlated (e.g., high humidity and high cloud cover co-occur). This assumption is evaluated in Section 7 by comparison with logistic regression[4].

4.2 Logistic Regression

As a discriminative benchmark, a logistic regression classifier was also trained. Rather than modelling the class-conditional likelihoods $P(\mathbf{x} \mid c)$, logistic regression directly models the posterior probability[4]:

$$P(\text{rain} \mid \mathbf{x}) = \frac{1}{1 + \exp(-(\mathbf{w}^T \mathbf{x} + b))} \tag{7}$$

where $\mathbf{w} \in \mathbb{R}^{18}$ is a weight vector and b is a bias term. The model makes no conditional independence assumption. Implementation used `sklearn.linear_model.LogisticRegression` with `class_weight="balanced"` to account for class imbalance.

5 Statistical Method

5.1 Likelihood

Under the conditional independence assumption, the joint likelihood factorises over features, and each feature is modelled as a univariate Gaussian. The class-conditional log-likelihood is therefore [4]:

$$\ln \mathcal{L}(c \mid \mathbf{x}; \boldsymbol{\theta}) = \sum_{j=1}^{d} \left[-\frac{(x_j - \mu_{jc})^2}{2\sigma_{jc}^2} - \frac{1}{2} \ln(2\pi\sigma_{jc}^2) \right] \tag{8}$$

5.2 Priors

The class priors are estimated from the training data as the empirical class frequencies:

$$P(c) = \frac{N_c}{N_{\text{train}}} \tag{9}$$

where N_c is the number of training days with label c and N_{train} is the total number of training days. The resulting priors are $P(\text{rain}) \approx 0.350$ and $P(\text{dry}) \approx 0.650$, consistent with the climatological rain frequency at Schiphol.

6 Inference Procedure

6.1 Gaussian Naive Bayes

Model parameters μ_{jc} and σ_{jc}^2 are estimated by maximum likelihood estimation (MLE) on the training set. For a Gaussian, the MLE reduces to the sample mean and variance:

$$\hat{\mu}_{jc} = \frac{1}{N_c} \sum_{i:\, y_i = c} x_{ij}, \qquad \hat{\sigma}_{jc}^2 = \frac{1}{N_c} \sum_{i:\, y_i = c} (x_{ij} - \hat{\mu}_{jc})^2 \tag{10}$$

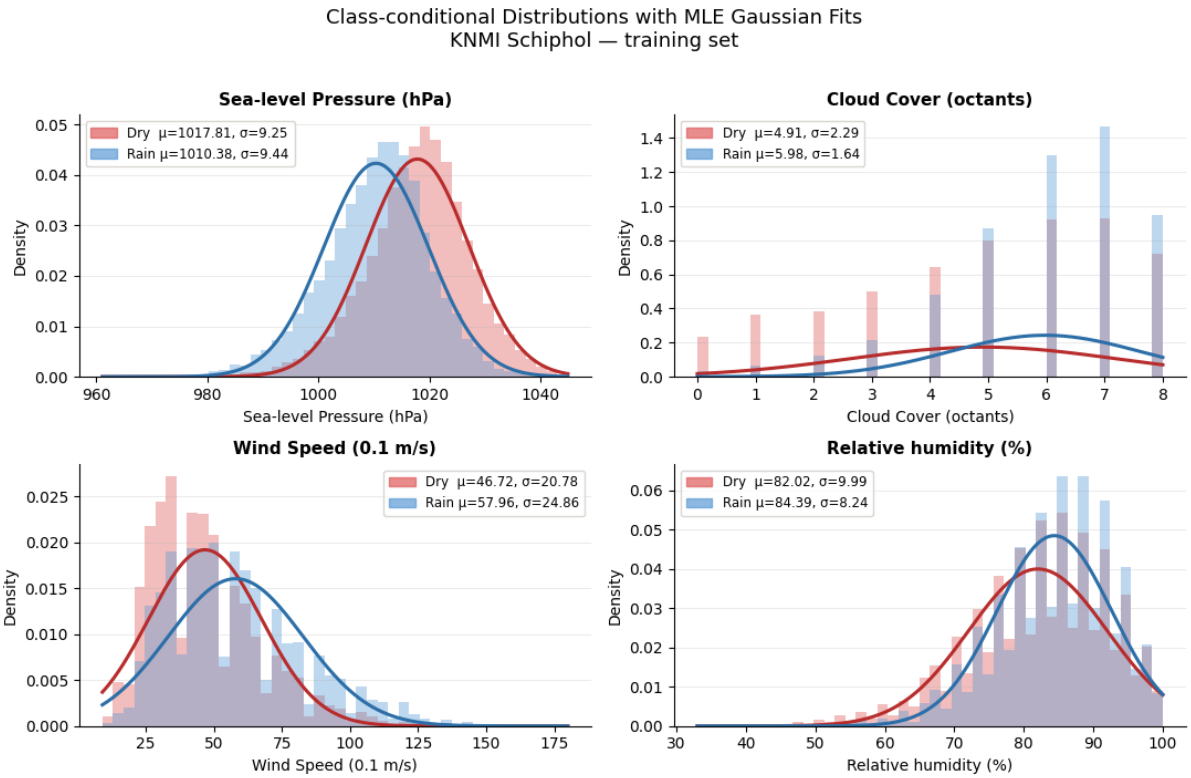

Figure 1: Class-conditional distributions for four selected predictors, with MLE Gaussian fits overlaid. Sea-level pressure and cloud cover show clear separation between classes. Sunshine duration exhibits a right-skewed distribution for dry days, indicating a departure from the Gaussian assumption.

Classification proceeds by computing the log-posterior for each class and predicting the class with the higher value:

$$\hat{y}_i = \mathbf{1}\Big[\ln \hat{\mathcal{L}}(\text{rain} \mid \mathbf{x}_i) + \ln P(\text{rain}) > \ln \hat{\mathcal{L}}(\text{dry} \mid \mathbf{x}_i) + \ln P(\text{dry})\Big] \tag{11}$$

6.2 Logistic Regression

The features were first standardised to zero mean and unit variance using `StandardScaler`, fitted on the training set only and then applied to the test set. A logistic regression classifier was then trained using `sklearn.linear_model.LogisticRegression` with `class_weight="balanced"` to account for the class imbalance between rain and dry days, and `max_iter=1000` to ensure convergence. Classification was performed by predicting the class with the higher posterior probability.

7 Results

7.1 Classification Performance

Table 2 summarizes the performance of the Gaussian Naive Bayes and logistic regression classifiers on the test set, together with the naive baseline (always predict dry).

Model	Accuracy	Rain Recall	Rain Precision	Rain F1
Naive baseline (always dry)	0.65	0.00	–	–
Gaussian Naive Bayes	0.70	0.64	0.58	0.60
Logistic Regression	0.68	0.76	0.54	0.63

Table 2: Classification performance on the held-out test set. All models use a chronological 80/20 train–test split.

7.2 Comparison of Models

Logistic regression and Gaussian Naive Bayes achieve comparable performance (Table 2). This is consistent with the theoretical result of Ng and Jordan [5], who show that the two classifiers are equivalent in the limit of infinite data when the conditional independence assumption holds, but that logistic regression has lower asymptotic error when it does not. The close results here suggest that feature correlations carry little additional information for this prediction task beyond what the Gaussian distributions already encode.

8 Discussion

The primary limitation of this approach is the single-day observational snapshot. The model uses only features from day t to predict rain on day $t + 1$, ignoring longer-term atmospheric trends. Multi-day pressure trends are only partially captured by the single delta feature `dPG`.
The Gaussian assumption for each feature may be violated. Variables such as sunshine duration (`SQ`) and wind speed (`FG`) have non-Gaussian, right-skewed distributions. A more flexible density estimator could better capture these marginals.
The class imbalance (35% rain days) causes the model to favour the dry class, as reflected in the lower rain precision compared to dry precision. Techniques such as adjusted decision thresholds could improve rain recall at the cost of dry-day accuracy.
Finally, the dataset is noted by KNMI to be inhomogeneous due to station relocations and changes in measurement methodology [3].

9 Possible Extensions

Several extensions could improve model performance:

- **Lag features**: including observations from days $t - 1$ and $t - 2$ would capture multi-day synoptic trends more faithfully.
- **Non-Gaussian likelihoods**: kernel density estimation or log-normal models for skewed variables such as wind speed and sunshine duration.
- **Seasonal stratification**: fitting separate Gaussian parameters per season, since the meteorological regime differs substantially between summer (convective) and winter (frontal) rainfall.
- **Threshold sensitivity analysis**: varying the rain threshold (currently 1 mm) and examining its effect on class balance and model performance.

10 Conclusion

This project demonstrated that next-day precipitation at Schiphol can be predicted with 70% accuracy using a Gaussian Naive Bayes classifier applied to 18 same-day meteorological features,

representing a 5 percentage point improvement over the naive climatological baseline. A rain recall of 64% shows that the model successfully identifies the majority of rainy days. Comparison with a logistic regression classifier revealed close performance, suggesting that the conditional independence assumption of the Naive Bayes model is approximately satisfied in this dataset. The results indicate that the primary limit on predictability is the single-day observational window rather than model complexity or distributional assumptions.

References

[1] John M. Wallace and Peter V. Hobbs. *Atmospheric Science: An Introductory Survey.* 2nd ed. Amsterdam: Academic Press, 2006.

[2] Roger G. Barry and Richard J. Chorley. *Atmosphere, Weather and Climate.* 9th ed. London: Routledge, 2009.

[3] Royal Netherlands Meteorological Institute (KNMI). *Daily weather observations, station Schiphol (240).* `https://www.knmi.nl/nederland-nu/klimatologie/daggegevens`. Accessed: 2026. 2024.

[4] Željko Ivezić et al. *Statistics, Data Mining, and Machine Learning in Astronomy: A Practical Python Guide for the Analysis of Survey Data.* Princeton University Press, 2020.

[5] Andrew Y. Ng and Michael I. Jordan. "On Discriminative vs. Generative Classifiers: A Comparison of Logistic Regression and Naive Bayes". In: *Advances in Neural Information Processing Systems.* Vol. 14. MIT Press, 2002.

A Additional Figures

B Reproducibility

The analysis was implemented in Python 3 using the following libraries: `numpy`, `pandas`, `scipy.stats`, `scikit-learn`, and `matplotlib`. The dataset `etmgeg_240.txt` was obtained directly from the KNMI website. All code and the raw data file are available at: github.com/stefanDorde/ilovegit.

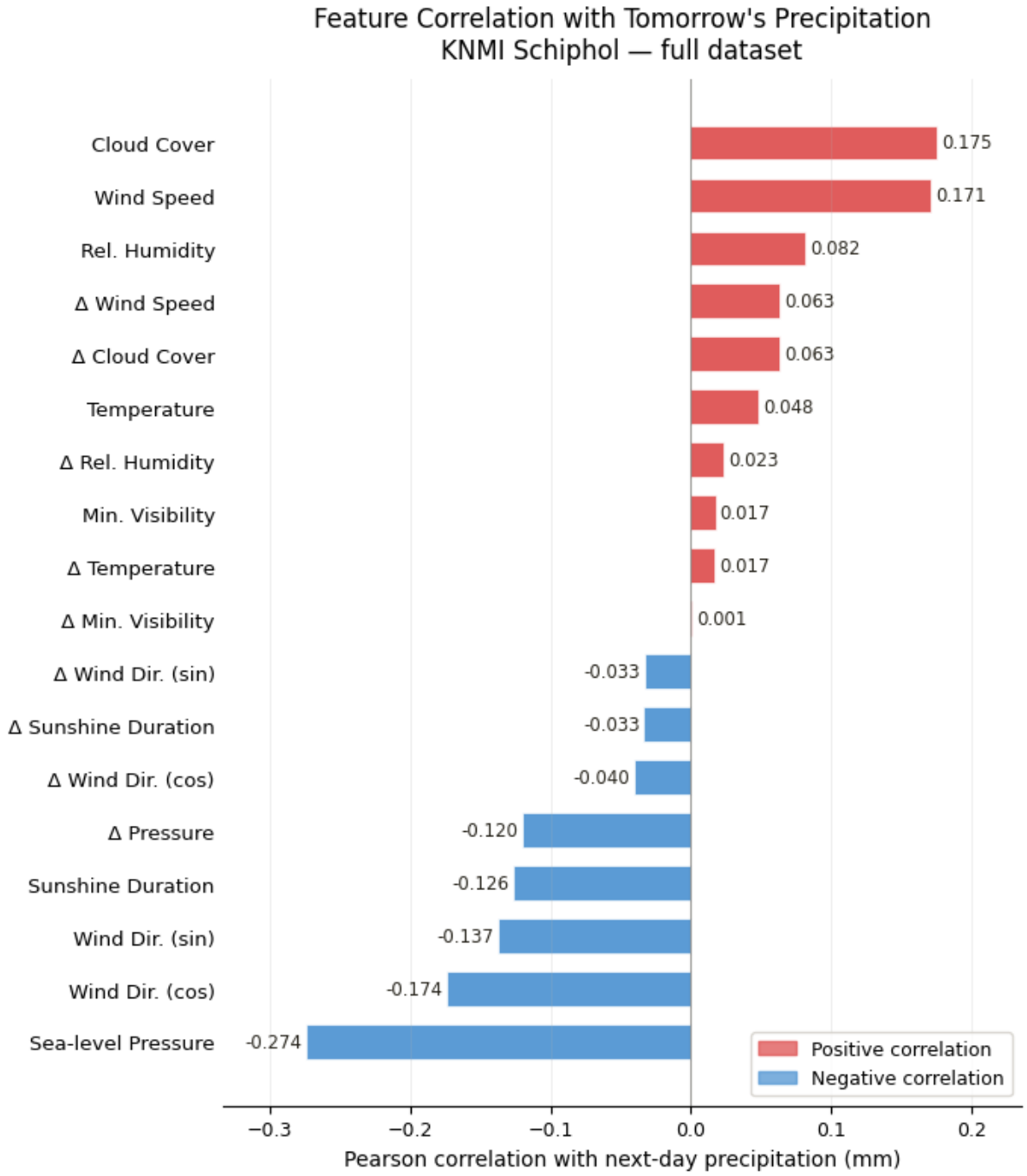

Figure 2: Pearson correlation coefficients between the 18 meteorological predictors and next-day precipitation (mm) at KNMI station Schiphol, computed over the full dataset. Level features generally show stronger linear associations than their first-difference counterparts (Δ), with sea-level pressure ($r = -0.27$) and cloud cover ($r = +0.18$) being the most informative individual predictors.